

COMP 425/COMP6341: Computer Vision (Winter 2026) Department of Computer Science and Software Engineering	
Course Instructor: Yang Wang yang.wang@concordia.ca (please refer to the “Other information” section of the syllabus on how to use email for communicating with the instructor)	
Office Hours: See information in Moodle	
Labs: Please see your class schedule for details Lab Demonstrators: see information in Moodle	
Course Calendar Description: <i>This course introduces basic techniques and concepts in computer vision including image formation, grouping and fitting, geometric vision, recognition, perceptual organization, and the state-of-the art software tools. Students learn fundamental algorithms and techniques, and gain experience in programming vision-based components.</i>	
Prerequisites: N/A Co-requisites: N/A	
Specific Knowledge and Skills Needed for this Course: Students taking this course are expected to have sufficient knowledge of the following topics. Should you have difficulties in any of these topics, you are strongly encouraged to review them before the DNE deadline. • good math background, including calculus, linear algebra, probability and statistics; • programming skills, data structures and algorithms; • (desirable) coursework or experience on machine learning; <i>Some homework assignments involve programming in Python using various libraries. Previous experience with Python is not required, although students are expected to learn the language on their own.</i>	
Course materials Required Textbook: N/A Optional Books: • D. Forsyth and J. Ponce, “Computer Vision: A Modern Approach”, 2nd edition, Prentice Hall, 2011. • R. Szeliski, “Computer Vision: Algorithms and Applications”, Springer, 2010. • R. Hartley and A. Zisserman, “Multiple View Geometry in Computer Vision”, Cambridge University Press, 2004. • C. Bishop, “Pattern Recognition and Machine Learning”, Springer, 2007. • I. Goodfellow, Y. Bengio and A. Courville, “Deep Learning”, http://www.deeplearningbook.org	
Grading Scheme • COMP425 ○ Assignments: 15%	

- Midterm exam: 30%
- Final exam: 55%
- COMP6341
 - Assignments: 10%
 - Midterm exam: 25%
 - Final exam: 50%
 - Project: 15% (including 1% for proposal)

Tentative Course Schedule

The following is a tentative list of topics to be covered (not necessarily in this order), subjective to minor change at the discretion of the instructor.

- Week1: introduction, image filters
- Week2: textures, frequency
- Week3: template matching, edge, keypoints
- Week4: fitting
- Week5: alignment
- Week6: image homography, image recognition
- Week7: midterm (in-class)
- Week8: reading week (no class)
- Week9: neural network
- Week10: ConvNet
- Week11: object detection
- Week12: dense prediction, RNN, attention, transformer
- Week13: advanced topics

Other information

- Communication

*Questions regarding course material should be posted in the discussion forum on Moodle. Students are encouraged to try to answer each others' questions if the instructor or TAs have not been able to answer immediately. **Emails should only be reserved for personal/confidential matters. When emailing the instructor or the TA, please follow these instructions to ensure that we do not miss your email and to help you receive a faster response: (1) use a meaningful subject; (2) send the email from your Concordia email account; (3) include your full name and student ID in the email; (4) include "comp425/6341" as part of the email subject. This will automatically put your email in a special folder in the instructor's email inbox. Otherwise, your email will be mixed with many other emails in the inbox and is more likely to get overlooked.***

If the instructor is unresponsive then he is likely overwhelmed with emails and messages and waiting for a block of time to work through them all and reset the "queue"; in that case the best way to communicate may be dropping by during office hours.

- Labs

This course has scheduled labs, however lab attendance is optional. The labs are mainly used as TA office hours where students can get help on assignments and course materials from the TAs.

- *Assignments*

There will be 2-3 homework assignments in this course. Each assignment will carry equal weight. Each assignment may involve a mix of written and/or programming questions. The dates for assignments will be determined based on the progress of the course materials.

Late Policy: *Students are allowed 3 grace days in total to use at their own discretion over the semester for the assignments. At most 2 grace days can be used for a particular assignment (e.g. you can use 2 grace days for HW1 and 1 grace days for HW2. But you can not use 3 grace days for HW1). Late days are counted from the time an assignment is due and is rounded up to the next whole day. For example, if an assignment is due on Friday 3:30pm, and is submitted on Saturday 6:00pm, 2 grace days will be used. Late assignments will not be accepted after either the 3 grace days are exhausted or 2 days past the due time of a particular assignment.*

- *Examinations*

Midterm exam: A midterm exam will be held on Feb 24, 2026. The midterm will be in the form of in-class paper-based exam. There will be no alternate or make-up exams. If you have a valid reason for missing the midterm (with proper documentation), your midterm weight will be shifted toward the final exam.

Final exam: A final exam will be held during the examination period at the end of the term. The exact date will be determined by the Exams Office. You must remain available during the exam period until all your exam obligations have been fulfilled.

- *Project*

COMP6341 students are required to complete a course project. Each project should be done in a group of 3-4 students. The project is an opportunity to explore a topic in-depth of your choosing. Each project group is required to submit a one-page proposal and a final report.

Graduate Attributes:

The following is the list of graduate attributes (skills) that students use, learn and/or apply throughout the term.

- *Knowledge Base (Developed/Intermediate): This course provides comprehensive knowledge of computer vision concepts and methods.*
- *Problem Analysis (Applied/Advanced): Emphasis is on applying theoretical knowledge to analyze real-world visual data and problems.*
- *Design (Developed/Intermediate): Focus on designing robust algorithms and systems for computer vision applications.*

- *Use of Tools (Introduced/Introductory): Introduction to, and practical experience with, essential computer vision tools and libraries.*

Course Learning Outcomes (CLOs):

By the end of this semester, students are expected to master the following engineering concepts.

- *Students will gain a deep understanding of key principles and techniques in computer vision.*
- *Students will develop the capability to analyze and interpret data from visual sources to solve complex problems in computer vision.*
- *Students will design and implement algorithms for common computer vision tasks.*
- *Students will acquire proficiency in using computer vision libraries and tools.*

Health and Safety Guidelines

All health and safety rules specific to this course can be found in the lab manual. General health and safety instructions and available health and safety trainings can be found at:

[Safety Programs - Concordia University \(https://www.concordia.ca/campus-life/safety/general-safety.html\)](https://www.concordia.ca/campus-life/safety/general-safety.html)

On Campus Resources

Please visit [Student services at Concordia University](#) for the services available Gina Cody School students.